SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Architecture

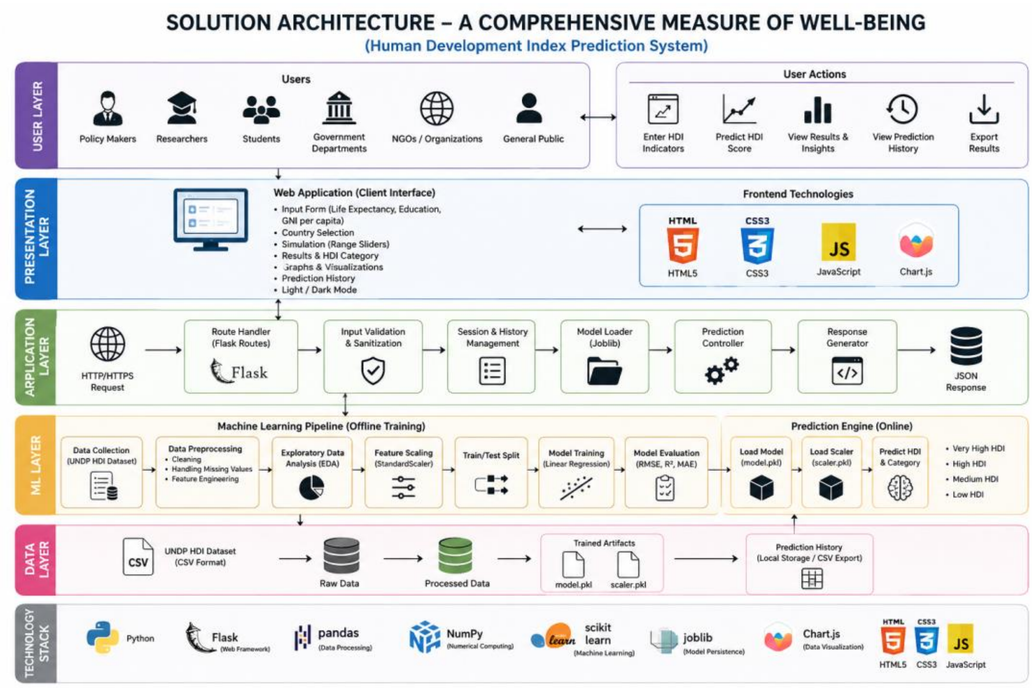
Date	**04 July 2026**
Team ID	
Project Name	**A Comprehensive Measure of Well-Being**
Maximum Marks	**4 Marks**

Solution Architecture:

The Solution Architecture for the **Human Development Index (HDI) Prediction System** provides an intelligent framework for predicting a country's Human Development Index using Machine Learning. Its goals are to:

1. Develop an accurate HDI prediction system using historical Human Development Index data.
2. Define the workflow between the dataset, Machine Learning model, and Flask web application.
3. Process user inputs and generate real-time HDI predictions and development classification.
4. Support scalable cloud deployment for reliable and accessible prediction services.

Example - Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Allows users to enter HDI indicators, submit requests, and view prediction results.	HTML, CSS, JavaScript
API Gateway	Receives user input, validates data, loads the model, and returns HDI predictions.	Flask, Python, Joblib
Prediction Engine	Predicts the HDI score and development category using the trained model.	Scikit-learn, Python, Linear Regression, Joblib
Database / Dataset	Stores the HDI dataset and supports model training and preprocessing.	Pandas, NumPy, Scikit-learn, UNDP HDI Dataset (CSV)